Phuong Doan

Senior AI Engineer

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Summary

Software engineer with 6+ years shipping production apps, including 2 years as a team lead, now building LLM, retrieval, and agent systems. Owns the code-intelligence, tool-admission, and vendor-gateway services of an internal AI platform used by ~100 people across engineering, marketing, product, and leadership; author of an MCP server on PyPI and the official MCP Registry.

Experience

Axon Active, Software Engineer (iOS → AI Platform)
Ho Chi Minh City · Hybrid
Tech stack: Java 21 / Quarkus · Python / FastAPI · Google ADK · TypeScript MCP · Vertex AI · pgvector · GKE · Cloud Build · Swift / SwiftUI
July 2024 – present
2 years 4 months

- Since 2026-06, core engineer on the company's **internal AI agent platform**: agents, tools, knowledge, and code search delivered to ~100 users' IDEs via MCP.
- Built the **code-intelligence service** (SCIP + Zoekt, hybrid search, call/type hierarchy, blast radius) indexing 11 pilot repos (~30k symbols in the largest) on free OSS, avoiding a ~$2.2k/month vendor tool.
- Designed the **tool admission gate** every agent tool call passes through: fail-open catalog/governance, fail-closed enforcement, full audit trail; verified with ArchUnit contracts and mutation testing.
- Shipped **per-user OAuth gateways for Jira/Confluence and Bitbucket** (OAuth 2.0 3LO, KMS envelope encryption, SSRF allowlists), unblocking agents on company data.
- Replaced the MCP server's fixed tool list with **discover / executeRead / executeWrite** meta-tools and hardened session expiry, so IDE agents reach the full catalog reliably.
- Earlier: built a UIKit + SwiftUI **design-system library** adopted by 3+ teams, cutting UI development time ~40% on the ePost iOS app.

S3 Corp, Software Engineer (iOS)
Ho Chi Minh City · Remote
Tech stack: Swift · RxSwift · Coordinator + MVVM · XCTest · WebSocket
Dec 2023 – June 2024
7 months

- Delivered feature screens on Coordinator + MVVM with RxSwift, integrating REST, APNs push, and real-time WebSocket connections.
- Reached **90% unit-test coverage** at the ViewModel layer with XCTest and RxTest.

ZILLEARN, iOS Engineer
Ho Chi Minh City · On-site
AI-powered learning platform for schools and enterprises. *Tech stack: Swift · tvOS · AVFoundation · Fastlane · Bitrise*
May 2022 – Dec 2023
1 year 8 months

- Led the refactor to Coordinator + MVVM across iOS and tvOS, and built the tvOS course and video-playback experience with AVFoundation.
- Set up CI/CD with Fastlane and Bitrise for automated Firebase App Distribution and TestFlight releases.

Windscribe, Software Engineer (iOS), Team Lead — Remote (Canada)
VPN provider. *Tech stack: Swift · OAuth 2.0 / PKCE · Keychain · Network Extension · Push extensions* — Apr 2020 – Mar 2022 — 2 years

- Led a team of 2–3 on the production VPN app, owning planning, code review, and feature strategy with the CEO and CTO.
- Implemented OAuth 2.0 with PKCE, AES credential encryption via Keychain, and zero-knowledge VPN tunnel key management.
- Built session-state and refresh-token handling, deep links, and a Notification Service Extension for rich push.

Selected Projects & Open Source

jarvis — local-first code-intelligence MCP server
Python · LanceDB · tree-sitter · SCIP · Zoekt — github.com/jarvis-intelligence/jarvis · PyPI jarvis-mcp · official MCP Registry

- Hybrid retrieval fusing dense vectors, Zoekt lexical search, and SCIP symbols via reciprocal rank fusion, so NL questions and exact-symbol lookups share one index.
- AST-aware chunking with tree-sitter, model-pinned embeddings, and atomic index publishing (zero-downtime reindex); shipped as compiled wheels with one-command plugins for Claude Code, Cursor, and Codex.

Stress AI Coach — iOS + watchOS app with streaming AI coach
Swift · HealthKit · StoreKit · Deno/Hono · SSE · OpenRouter · Postgres — live on the App Store · backend

- Privacy-first stress scoring from HealthKit with a streaming AI coach; raw health data never leaves the device, only derived scores reach the model.
- Self-owned backend: SSE proxy over a 4-model fallback chain, crash-safe credit metering, Firebase auth, offline StoreKit receipt verification, Postgres on Dokploy.

LLM gateway & agent observability
FastAPI · React · TypeScript · SSE — github.com/phuongddx/llm-gateway

- OpenAI-compatible gateway routing to 500+ models behind one key, capturing TTFT, latency, and token usage per request to pick models on cost and speed.
- Agent-run control plane with DAG trace viewer, run comparison, and per-model circuit-breaker states to surface degrading providers early.

Open-source contributions

- **scip-swift** — extended SCIP indexing to Swift via SourceKit-LSP; contributed a typeHierarchy fix upstream to scip-code/scip. **OmniRoute** — 2 merged PRs adding SSE streaming for GLM-5.3 effort tiers. **spm-cache** — SPM xcframework cache proxy with Homebrew tap and GitHub Action.

Education

Ho Chi Minh City University of Technology, Software Engineering — 2013 – 2017